Supplementary Material

Sup. Table 1. PRISMA Checklist [1]

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	Page 1; lines 1-2
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	Page 2; lines 2-18
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	Pages 3 and 4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	Page 4; lines 15-22
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	Page 5; lines 1-3
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	Page 5; lines 4-13
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	Page 5; line 14-18
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Page 5; line 19-25 Supplementary Table 2
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	Page 3; line 25, 26 Page 4; lines 1-7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	Page 5; lines 19-25
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	Page 7; lines 5-13
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	Page 6; lines 7-14

Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	Page 6; lines 15-25
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	Page 6; lines 15-25
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	Page 7; lines 1-4
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	Page 7; lines 1-4
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	Page 7; line 15, 17
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Page 7; lines 18-26; Page 8; lines 1-14; Tables 1-3
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Sup. Table 3
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	Figure 2 and 3 Page 8; lines 21-26 Page 9; lines 1-12
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Figure 2 and 3
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	Sup. Figure 1 Page 8; lines 16-20
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Figure S1
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	Pages 9; lines 14-17
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	Page 11; lines 13-17
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	Page 11; lines 18-23
FUNDING			

Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	Page 11; lines 24-25

Sup. Table 2. Search strategies and the number of records according to different electronic database

Search strategy	**Database**	**Num. of records**
((((((((((obesity[MeSH Terms]) OR (central obesity[MeSH Terms])) OR (overweight[MeSH Terms])) OR (body mass index[MeSH Terms])) OR (abdominal obesity[MeSH Terms])) OR (overweight[MeSH Terms])) OR (waist circumference[MeSH Terms])) OR (ratio, waist to hip[MeSH Terms])) OR (body fat distribution[MeSH Terms])) OR (abdominal fat[MeSH Terms])) OR(body composition[MeSH Terms])) AND ((((((((((sedentary behav*[MeSH Terms]) OR (screen time[MeSH Terms])) OR (sitting time[MeSH Terms])) OR (sitting time[MeSH Terms])) OR (television view*[MeSH Terms])) OR (watching television[MeSH Terms])) OR (computer use[MeSH Terms])) OR (internet use[MeSH Terms])) OR (smart phone[MeSH Terms])) OR (video game*[MeSH Terms])) OR (electronic game*[MeSH Terms])) OR (depress*[MeSH Terms]))) AND ((((((((((Child*[MeSH Terms]) OR (children[MeSH Terms])) OR (teen*[MeSH Terms])) OR (adolescent[MeSH Terms])) OR (boy*[MeSH Terms])) OR (girl[MeSH Terms])) OR (all, childhood[MeSH Terms])) OR (pediatric*[MeSH Terms])) OR (youth[MeSH Terms])) OR (teenager[MeSH Terms])) OR (toddler*[MeSH Terms]))	PubMed	2121
	Scopus	2016
	Embase	2154

Sup. Table 3. Agency for Healthcare Research and Quality (AHRQ) checklist to assess quality of the cross-sectional studies

ARHQ Methodology Checklist items for Cross-Sectional study	Hoffmann B [2]	Tsuchiya M [3]	Kelishadi R [4]	Alghadir AH [5]	Al-Agha AE [6]	Stamatakis E [7]	Hardy LL [8]	Cheng L [9]	Elizondo-MontemayorL [10]
1) Define the source of information (survey, record review)	⊕	⊕	⊕	⊕	⊕	⊕	⊕	⊕	⊕
2) List inclusion and exclusion criteria for exposed and unexposed subjects (cases and controls) or refer to previous publications	⊕	⊕	⊕	-	⊕	⊕	U	⊕	-
3) Indicate time period used for identifying patients	-	⊕	⊕	⊕	⊕	⊕	⊕	⊕	⊕
4) Indicate whether or not subjects were consecutive if not population-based	⊕	-	-	-	-	-	⊕	-	-
5) Indicate if evaluators of subjective components of study were masked to other aspects of the status of the participants	U	⊕	U	-	-	-	U	-	-
6) Describe any assessments undertaken for quality assurance purposes (e.g., test/retest of primary outcome measurements)	-	-	-	-	-	-	-	⊕	-
7) Explain any patient exclusions from analysis	-	⊕	⊕	⊕	⊕	⊕	⊕	⊕	
8) Describe how confounding was assessed and/or controlled.	⊕	⊕	⊕	⊕	-	⊕	⊕	⊕	⊕
9) If applicable, explain how missing data were handled in the analysis	-	⊕	⊕	⊕	⊕	⊕	⊕	⊕	⊕
10) Summarize patient response rates and completeness of data collection	⊕	⊕	⊕	-	⊕	⊕	⊕	⊕	⊕
11) Clarify what follow-up, if any, was expected and the percentage of patients for which incomplete data or follow-up was obtained	-	-	-	-	-	-	-	-	-
Total score	5	8	7	5	5	7	7	8	5

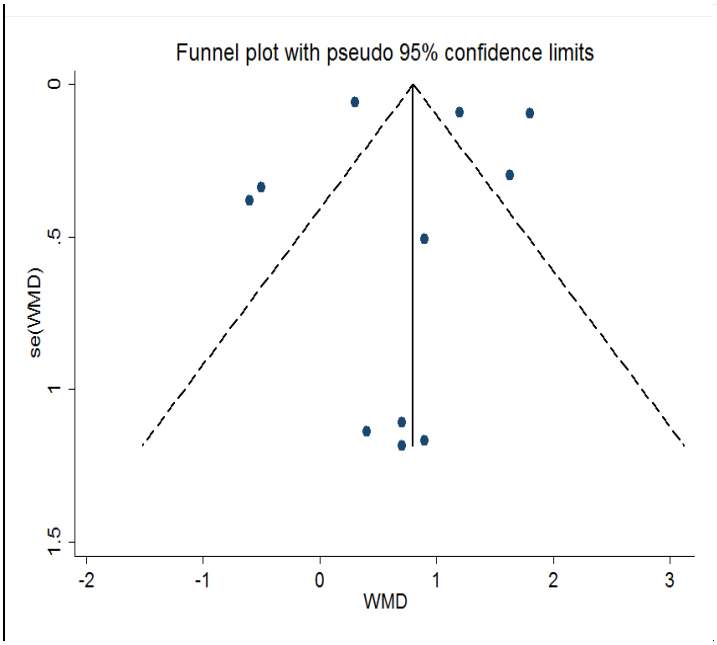
Sup. Table 3. Cont'd

ARHQ Methodology Checklist items for Cross-Sectional study	Werneck AO [11]	Suchert V [12]	Börnhorst C [13]	Canan F [14]	Appelhans BM [15]	Ghavamzadeh S [16]	Delong AJ [17]	**Safiri S [18]**	**Christofaro DGD [19]**
1) Define the source of information (survey, record review)	⊕	⊕	⊕	⊕	⊕	⊕	⊕	⊕	⊕
2) List inclusion and exclusion criteria for exposed and unexposed subjects (cases and controls) or refer to previous publications	-	⊕	⊕	⊕	⊕	⊕	⊕	⊕	⊕
3) Indicate time period used for identifying patients	⊕	⊕	⊕	⊕	⊕	⊕	⊕	⊕	-
4) Indicate whether or not subjects were consecutive if not population-based	-	-	-	-	-	-		⊕	⊕
5) Indicate if evaluators of subjective components of study were masked to other aspects of the status of the participants	-	U	U	U	U	-		U	U
6) Describe any assessments undertaken for quality assurance purposes (e.g., test/retest of primary outcome measurements)	⊕	-	-	-	-	-		U	U
7) Explain any patient exclusions from analysis		⊕	⊕	⊕	⊕	⊕	⊕	⊕	U
8) Describe how confounding was assessed and/or controlled.	⊕	⊕	⊕	⊕	⊕	⊕	⊕	⊕	⊕
9) If applicable, explain how missing data were handled in the analysis	-	⊕	⊕	⊕	⊕	-	⊕	-	U
10) Summarize patient response rates and completeness of data collection	⊕	⊕	⊕	⊕	⊕	-	⊕	-	⊕
11) Clarify what follow-up, if any, was expected and the percentage of patients for which incomplete data or follow-up was obtained	-	-	-	-	-	-	-	U	-
	5	7	7	7	7	5	7	6	5

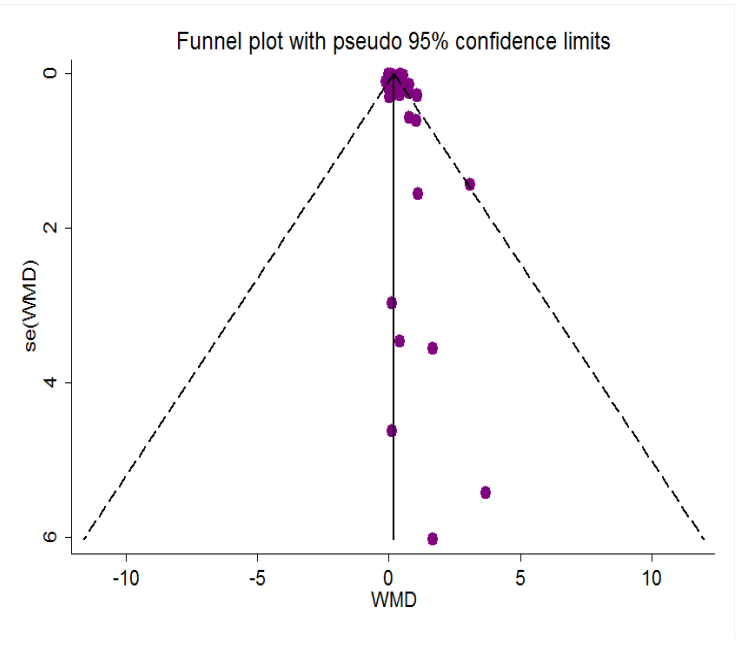
Sup. Table 3. Cont'd

ARHQ Methodology Checklist items for Cross-Sectional study	Asplund KM [20]	Hendrix KS [21]	Berentzen NE [22]	Shriver L [23]	Vrijkotte TGM [24]	Danielsen YS [25]	Kerkadi A [26]	Elgar FJ [27]	Fazah A [28]
1) Define the source of information (survey, record review)	⊕	⊕	⊕	⊕	⊕	⊕	⊕	⊕	⊕
2) List inclusion and exclusion criteria for exposed and unexposed subjects (cases and controls) or refer to previous publications	⊕	⊕	⊕	⊕	⊕	⊕	⊕	⊕	⊕
3) Indicate time period used for identifying patients	⊕	⊕	⊕	⊕	⊕	⊕	_	⊕	⊕
4) Indicate whether or not subjects were consecutive if not population-based	-	-	-	-	⊕	⊕	⊕	-	-
5) Indicate if evaluators of subjective components of study were masked to other aspects of the status of the participants	U	U	-	U	U	U	U	-	-
6) Describe any assessments undertaken for quality assurance purposes (e.g., test/retest of primary outcome measurements)	⊕	-	-	-	-	-	U	-	-
7) Explain any patient exclusions from analysis	⊕	-	⊕	⊕	⊕	⊕	_	⊕	⊕
8) Describe how confounding was assessed and/or controlled.	⊕	⊕	⊕	⊕	⊕	⊕	⊕	⊕	⊕
9) If applicable, explain how missing data were handled in the analysis	⊕	-	⊕	⊕	⊕	⊕	⊕	⊕	⊕
10) Summarize patient response rates and completeness of data collection	⊕	⊕	⊕	⊕	⊕	⊕	⊕	⊕	⊕
11) Clarify what follow-up, if any, was expected and the percentage of patients for which incomplete data or follow-up was obtained	-	-	-	-	-	-	⊕	⊕	⊕
	8	5	7	7	8	8	7	8	8

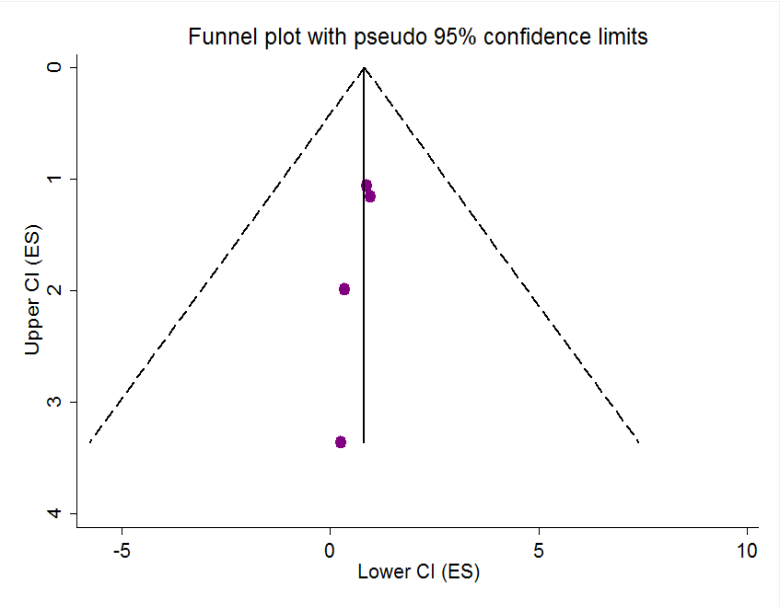
(A)



(B)



(C)



Sup. Figure 1. Begg's funnel plot (with pseudo 95% CIs) of the comparison of BMI between highest versus lowest screen time categories (A); the comparison of screen time (ST) between obese versus non-obese youth (B); odds of screen addiction among obese versus non obese youth (C).

References

1. **Moher D, Liberati A, Tetzlaff J, Altman DG**. 2009. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. Annals of internal medicine;**151(4)**:264-9.
2. **Hoffmann B, Kobel S, Wartha O, Kettner S, Dreyhaupt J, Steinacker JM**. 2019. High sedentary time in children is not only due to screen media use: a cross-sectional study. BMC pediatrics;**19(1)**:154.
3. **Tsuchiya M, Momma H, Sekiguchi T, Kuroki K, Kanazawa K, Watanabe M, et al.** 2017. Excessive game playing is associated with poor toothbrushing behavior among athletic children: A cross-sectional study in Miyagi, Japan. Tohoku Journal of Experimental Medicine;**241(2)**:131-8.
4. **Kelishadi R, Mozafarian N, Qorbani M, Maracy MR, Motlagh ME, Safiri S, et al.** 2017. Association between screen time and snack consumption in children and adolescents: The CASPIAN-IV study. Journal of Pediatric Endocrinology and Metabolism;**30(2)**:211-9.
5. **Alghadir AH, Gabr SA, Iqbal ZA**. 2016. Television watching, diet and body mass index of school children in Saudi Arabia. Pediatrics international : official journal of the Japan Pediatric Society;**58(4)**:290-4.
6. **Al-Agha AE, Nizar FS, Nahhas AM**. 2016. The association between body mass index and duration spent on electronic devices in children and adolescents in Western Saudi Arabia. Saudi medical journal;**37(4)**:436-9.
7. **Stamatakis E, Coombs N, Jago R, Gama A, Mourão I, Nogueira H, et al.** 2013. Associations between indicators of screen time and adiposity indices in Portuguese children. Preventive medicine;**56(5)**:299-303.
8. **Hardy LL, Denney-Wilson E, Thrift AP, Okely AD, Baur LA**. 2010. Screen time and metabolic risk factors among adolescents. Archives of Pediatrics and Adolescent Medicine;**164(7)**:643-9.
9. **Cheng L, Pohlabeln H, Ahrens W, Lauria F, Veidebaum T, Chadjigeorgiou C, et al.** 2020. Cross-sectional and longitudinal associations between physical activity, sedentary behaviour and bone stiffness index across weight status in European children and adolescents. International Journal of Behavioral Nutrition and Physical Activity;**17(1)**.
10. **Elizondo-Montemayor L, Ugalde-Casas PA, Serrano-González M, Cuello-García CA, Borbolla-Escoboza JR**. 2010. Serum 25-hydroxyvitamin D concentration, life factors and obesity in mexican children. Obesity;**18(9)**:1805-11.
11. **Werneck AO, Silva DR, Agostinete RR, Fernandes RA, Ronque ERV, Cyrino ES**. 2018. Social, behavioral and biological correlates of cardiorespiratory fitness according to sex, nutritional status and maturity status among adolescents. A cross-sectional study. Sao Paulo Medical Journal;**136(3)**:237-44.
12. **Suchert V, Hanewinkel R, Isensee B**. 2016. Screen time, weight status and the self-concept of physical attractiveness in adolescents. Journal of adolescence;**48**:11-7.
13. **Börnhorst C, Wijnhoven TMA, Kunešová M, Yngve A, Rito AI, Lissner L, et al.** 2015. WHO European Childhood Obesity Surveillance Initiative: Associations between sleep duration, screen time and food consumption frequencies. BMC public health;**15(1)**.
14. **Canan F, Yildirim O, Ustunel TY, Sinani G, Kaleli AH, Gunes C, et al.** 2014. The relationship between internet addiction and body mass index in Turkish adolescents. Cyberpsychology, Behavior, and Social Networking;**17(1)**:40-5.
15. **Appelhans BM, Fitzpatrick SL, Li H, Cail V, Waring ME, Schneider KL, et al.** 2014. The home environment and childhood obesity in low-income households: Indirect effects via sleep duration and screen time. BMC public health;**14(1)**.
16. **Ghavamzadeh S, Khalkhali HR, Alizadeh M**. 2013. TV viewing, independent of physical activity and obesogenic foods, increases overweight and obesity in adolescents. Journal of health, population, and nutrition;**31(3)**:334-42.
17. **Delong AJ, Larson NI, Story M, Neumark-Sztainer D, Weber-Main AM, Ireland M**. 2008. Factors associated with overweight among urban American Indian adolescents: Findings from project EAT. Ethnicity and Disease;**18(3)**:317-23.
18. **Safiri S, Kelishadi R, Qorbani M, Abbasi-Ghah-Ramanloo A, Motlagh ME, Ardalan G, et al.** 2015. Screen time and its relation to cardiometabolic risk among children and adolescents: The CASPIAN-III study. Iranian Journal of Public Health;**44(1)**:35-44.
19. **Christofaro DGD, De Andrade SM, Cardoso JR, Mesas AE, Codogno JS, Fernandes RA**. 2015. High blood pressure and sedentary behavior in adolescents are associated even after controlling for confounding factors. Blood pressure;**24(5)**:317-23.
20. **Asplund KM, Kair LR, Arain YH, Cervantes M, Oreskovic NM, Zuckerman KE**. 2015. Early childhood screen time and parental attitudes toward child television viewing in a low-income latino population attending the special supplemental nutrition program for women, infants, and children. Childhood Obesity;**11(5)**:590-9.
21. **Hendrix KS, Carroll AE, Downs SM**. 2014. Screen exposure and body mass index status in 2- to 11-year-old children. Clinical pediatrics;**53(6)**:593-600.
22. **Berentzen NE, Smit HA, Van Rossem L, Gehring U, Kerkhof M, Postma DS, et al.** 2014. Screen time, adiposity and

cardiometabolic markers: Mediation by physical activity, not snacking, among 11-year-old children. International Journal of Obesity;**38**:1317-23.
23. **Shriver LH, Harrist AW, Hubbs-Tait L, Topham G, Page M, Barrett A**. 2011. Weight status, physical activity, and fitness among third-grade rural children. The Journal of school health;**81(9)**:536-44.
24. **Vrijkotte TGM, Oostvogels AJJM, Stronks K, Roseboom TJ, Hof MHP**. 2020. Growth patterns from birth to overweight at age 5-6 years of children with various backgrounds in socioeconomic status and country of origin: the ABCD study. Pediatric obesity;**15(8)**.
25. **Danielsen YS, Júlíusson PB, Nordhus IH, Kleiven M, Meltzer HM, Olsson SJ, et al.** 2011. The relationship between life-style and cardio-metabolic risk indicators in children: the importance of screen time. Acta paediatrica (Oslo, Norway : 1992);**100(2)**:253-9.
26. **Kerkadi A, Sadig AH, Bawadi H, Thani AAMA, Chetachi WA, Akram H, et al.** 2019. The relationship between lifestyle factors and obesity indices among adolescents in Qatar. International journal of environmental research and public health;**16(22)**.
27. **Elgar FJ, Moore L, Roberts C, Tudor-Smith C**. 2004. Sedentary behaviour, physical activity and weight problems among adolescents in wales. Psychology and Health;**19(SUPPL. 1)**:48-9.
28. **Fazah A, Jacob C, Moussa E, El-Hage R, Youssef H, Delamarche P**. 2010. Activity, inactivity and quality of life among Lebanese adolescents. Pediatrics International;**52(4)**:573-8.